

6.2 Deformation: Elastic & Plastic Behaviour

Question Paper

Course	CIEA Level Physics
Section	6. Deformation of Solids
Topic	6.2 Deformation: Elastic & Plastic Behaviour
Difficulty	Hard

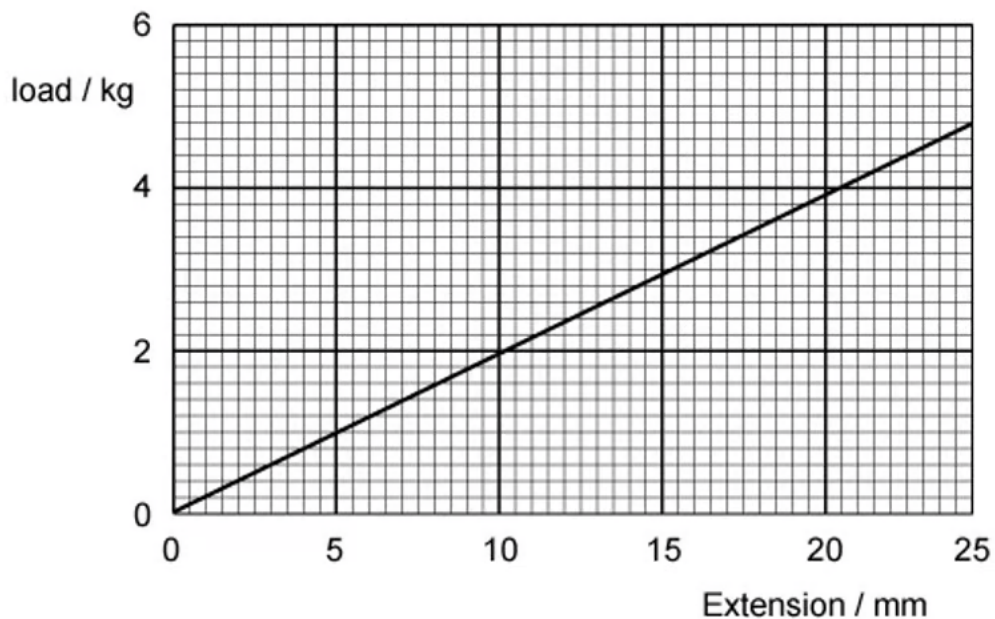
Time allowed: 10

Score: /5

Percentage: /100

Question 1

A wire undergoes elastic deformation; the graph below shows the load-extension graph.



How much work is done on the wire to increase the extension from 10 mm to 20 mm?

- A. 0.37 J
- B. 0.28 J
- C. 0.184 J
- D. 0.028 J

[1 mark]

Question 2

A material with elastic properties that obeys Hooke's law has a Young modulus E and has a tensile stress of S applied.

What is the expression for the elastic energy stored per unit volume of the material?

A. $\frac{E}{2S^2}$

B. $\frac{S^2}{E}$

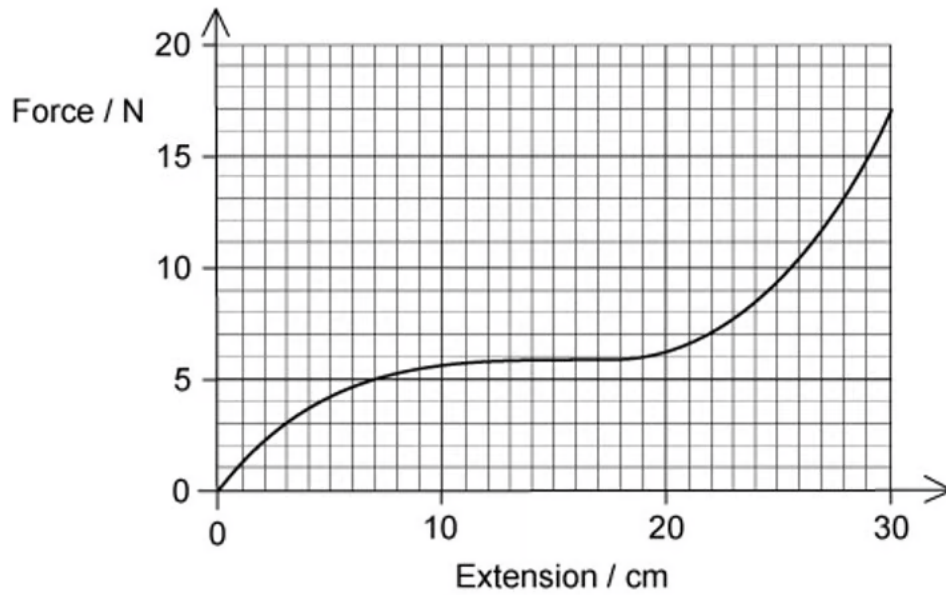
C. $\frac{S^2}{2E}$

D. $\frac{2E}{S^2}$

[1 mark]

Question 3

A graph was drawn of the effect on the length of applying a force to a rubber cord.



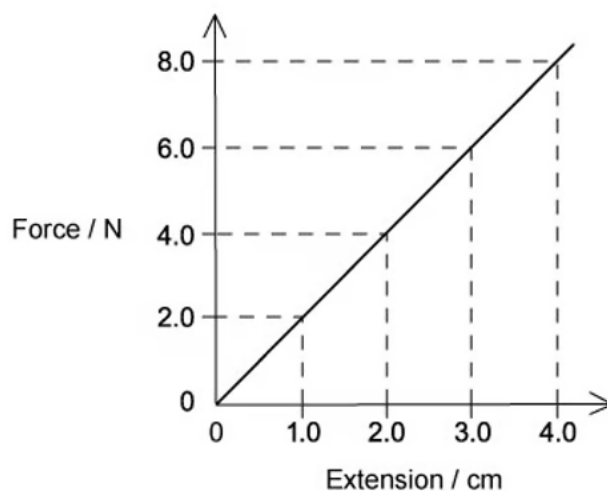
What is the maximum strain energy in this deformed rubber cord?

- A. 200 J
- B. 5.1 J
- C. 2.5 J
- D. 1.9 J

[1 mark]

Question 4

The graph below shows how force changes the extension of a spring.



The length of the spring with no force applied was 1 cm.

What is the **increase** in the strain energy stored in the spring when its **length** is increased from 2.0 cm to 3.0 cm?

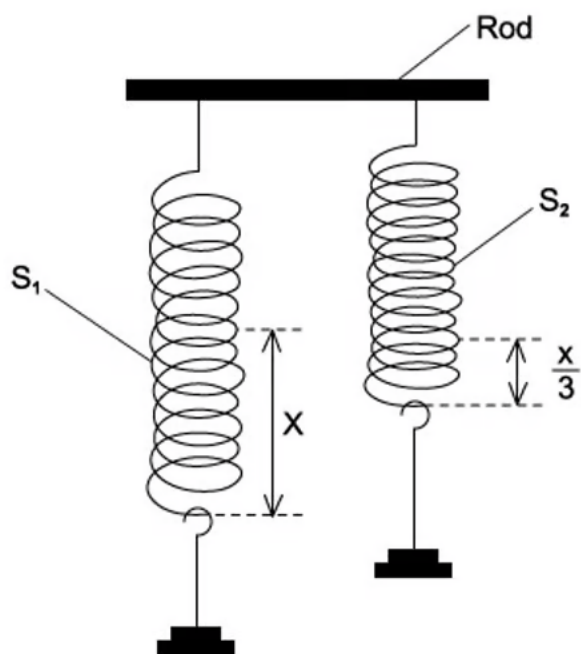
- A. 0.050 J
- B. 0.040 J
- C. 0.030 J
- D. 0.020 J

[1 mark]

Question 5

Two springs are attached to a rod, S_1 and S_2 . A load is applied to the bottom of S_1 , and it extends by x with an elastic potential energy in the spring of E .

The same load is applied to the bottom of S_2 . The extension of S_2 is $\frac{x}{3}$, as shown in the diagram



Which equation shows the elastic potential energy in S_2 ?

- A. $9E$
- B. $3E$
- C. $\frac{E}{3}$
- D. $\frac{E}{9}$

[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- Strain energy $= \frac{1}{2}Fx$
 - So, the area under the force-extension graph equals the work done
- The load that is applied needs to be converted from kg to N by multiplying by g
- The load is not constant as the extension is increasing, so the area under the line from 10 mm to 20 mm extension is needed to calculate the work done
 - Extension $= 20 - 10 = 10 \text{ mm} = 10 \times 10^{-3} \text{ m}$
 - Load at 10 mm $= 1.9 \text{ kg}$
 - Load at 20 mm $= 3.8 \text{ kg}$
- Work done $= \frac{1}{2}(1.9 + 3.8)(9.81)(10 \times 10^{-3}) = 0.279 \text{ J}$
 - Therefore, work is done on the wire is 0.28 J

2

The correct answer is **C** because:

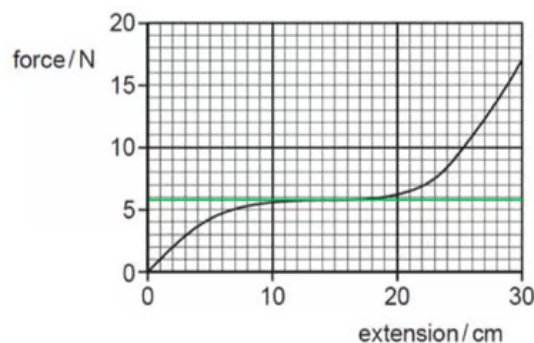
- Elastic energy $= \frac{1}{2}Fx$
- Elastic energy per unit volume $= \frac{Fx}{2V}$
- Volume $= AL$, so, substituting this into the elastic energy per unit volume equation:
 - Elastic energy per unit volume $= \frac{Fx}{2AL} = \frac{x}{2L} \left(\frac{F}{A} \right)$
 - Tensile stress S is equal to $= \frac{F}{A}$
 - So, the elastic energy per unit volume $= \frac{xS}{2L}$

- Strain is equal to $= \frac{x}{L}$
 - So, elastic energy per unit volume $= \frac{S}{2} \left(\frac{x}{L} \right)$
- Young modulus $E = \frac{\text{stress}}{\text{strain}}$
 - Therefore, strain $= \frac{x}{L} = \frac{S}{E}$
- Combining the volume and strain equations
 - Elastic energy per unit volume $= \frac{s}{2} \left(\frac{S}{E} \right) = \frac{S^2}{2E}$

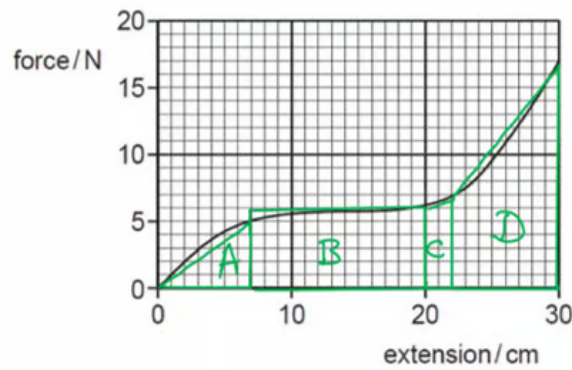
3

The correct answer is **D** because:

- The area under the force-extension graph gives the energy stored in the rubber cord
- Method 1: Draw a straight line on the graph to give the approximate mean force
 - Giving $6 \text{ N} \times 0.3 \text{ m} = 1.8 \text{ J}$



- Method 2: Split the area into a series of triangles and rectangles to work out the total area

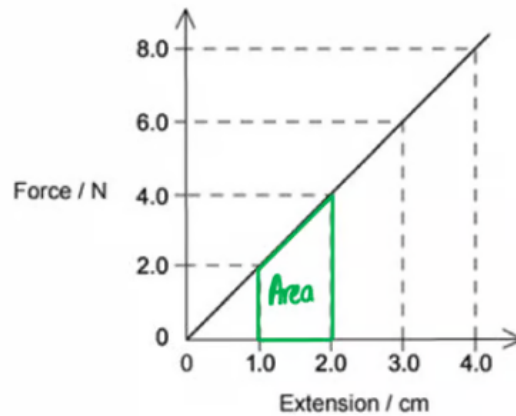


- ■ Area A = $\frac{1}{2} \times 0.07 \times 5 = 0.175 \text{ J}$
- Area B = $(0.20 - 0.07) \times 5.5 = 0.715 \text{ J}$
- Area C = $\frac{1}{2} \times (5.5 + 6) \times 0.02 = 0.115 \text{ J}$
- Area D = $\frac{1}{2} \times (6 + 17) \times (0.30 - 0.22) = 0.92 \text{ J}$
- Total area = $0.175 + 0.715 + 0.115 + 0.92 = 1.925 \text{ J}$
- The best estimate is option **D**

4

The correct answer is **C** because:

- The strain energy is the energy stored in an elastic body under loading
- The spring is elastic and obeying Hooke's law as the graph is proportional and goes through the origin
 - The spring has an original length of 1.0 cm
 - The strain energy is equal to the area under the force-extension graph.
- The question asks for the increase in length, not extension
- Original length = 1.0 cm
 - When length = 2.0 cm, extension = $2.0 - 1.0 = 1.0 \text{ cm}$ extension
 - When length = 3.0 cm, extension = $3.0 - 1.0 = 2.0 \text{ cm}$
- The strain energy can be calculated from the area under the graph between 1.0 cm and 2.0 cm, $1.0 \text{ cm} = 0.01 \text{ m}$



- Area of a trapezium $\frac{1}{2} \times \sum \text{parallel sides} \times \text{height}$
- Therefore, the strain energy stored is $= \frac{1}{2} \times (2.0 + 4.0) \times 0.01 = 0.030 \text{ J}$

5

The correct answer is **C** because:

- Elastic potential energy is the energy stored as a result of deformation of an elastic object
 - Elastic potential energy $= \frac{1}{2}Fx$
- The spring constant k of each of the springs is different as well as the extension x due to the load applied
- For spring S_1 :
 - $E = \frac{1}{2}Fx$
 - $F = \frac{2E}{x}$
- For spring S_2 :
 - $E_2 = \frac{1}{2}F\left(\frac{x}{3}\right) = \frac{1}{6}Fx$
- Since the load F applied to both springs is the same
 - $F = \frac{6E_2}{x}$

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 - $F = \frac{6E_2}{x}$
 - $F = \frac{2E}{x} = \frac{6E_2}{x}$
 - $E_2 = \frac{2E}{6} = \frac{E}{3}$

Therefore, if spring S_1 extends by x , then the energy stored in S_2 will be $\frac{E}{3}$